

PORT EDWARD, South Africa

WMO: 685870

Lat: 31.0622S

Lon: 30.2214E

Elev: 36

StdP: 14.68

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	53.0	54.6	31.9	26.4	61.3	35.8	30.8	60.5	24.1	65.8	21.3	65.3	5.3	320	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.4	82.2	75.5	81.0	74.9	80.0	74.2	77.8	79.9	76.3	79.1	75.1	78.5	9.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	142.0	79.2	75.5	133.6	78.1	74.0	127.0	77.5	41.4	79.9	39.9	79.1	38.7	78.5	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.6	18.8	16.5	DB	48.0	85.7	2.5	2.6	46.2	87.6	44.8	89.2	43.4	90.7	41.6	92.6	(4)
(5)			WB	41.9	77.8	1.5	1.7	40.9	79.0	40.0	79.9	39.2	80.9	38.1	82.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.0	74.2	74.9	73.6	70.3	67.6	64.5	63.4	64.6	66.0	67.2	69.7	72.2
	DBStd	4.87	2.76	2.67	2.69	3.22	2.81	2.69	2.70	3.06	3.32	3.22	3.23	3.05
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	205	0	0	0	4	9	39	62	43	27	17	4	0
	CDD50	6927	750	696	731	608	545	435	416	454	479	533	590	690
	CDD65	1657	285	276	266	162	89	24	13	32	56	85	144	225
	CDH74	5273	1285	1423	1051	348	127	30	22	22	27	42	221	675
	CDH80	230	67	92	31	5	5	1	3	1	2	0	6	17
Wind	WSAvg	7.7	7.7	7.1	7.6	7.3	7.0	6.8	7.0	8.0	8.5	8.9	8.6	8.0
Precipitation	PrecAvg	43.3	5.0	4.1	6.1	3.2	1.8	1.2	1.9	1.8	3.5	4.5	5.5	5.4
	PrecMax	66.9	12.1	10.1	15.3	6.3	3.9	5.8	6.7	5.8	9.2	9.4	9.9	8.7
	PrecMin	32.0	2.1	1.5	1.9	1.0	0.2	0.1	0.0	0.3	0.7	1.1	2.4	1.9
	PrecStd	10.1	2.4	2.1	3.3	1.7	1.1	1.4	2.1	1.5	2.1	2.2	2.3	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.5	84.0	82.4	80.7	79.9	77.0	76.5	76.5	76.4	77.0	79.6	81.8
		MCWB	78.0	78.5	74.6	73.4	66.1	61.9	60.4	65.1	67.6	69.2	72.1	76.5
	2%	DB	81.7	82.3	80.9	78.7	77.0	74.5	73.4	73.6	74.2	75.2	77.7	79.9
		MCWB	76.4	76.1	74.6	72.4	68.4	62.9	62.6	65.4	67.8	68.8	72.2	75.0
	5%	DB	80.6	81.2	79.9	77.4	75.2	72.8	71.7	72.1	73.0	73.8	76.4	78.7
		MCWB	75.3	75.3	73.9	71.0	67.2	62.6	62.9	64.8	66.9	67.8	71.4	73.8
	10%	DB	79.3	80.0	78.8	76.1	73.8	71.3	70.2	70.9	71.9	72.7	75.1	77.5
		MCWB	73.9	74.4	73.1	69.9	66.7	61.9	62.0	64.1	66.0	66.9	70.2	72.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.8	79.5	78.2	74.9	71.8	69.0	67.8	69.9	71.0	71.2	75.6	78.8
		MCDB	81.7	81.0	80.0	78.7	76.1	72.3	71.0	72.6	74.3	74.1	77.2	79.9
	2%	WB	77.7	77.4	76.4	73.5	70.0	66.3	65.9	67.9	69.6	69.9	73.7	76.4
		MCDB	80.4	80.0	78.8	77.4	74.8	72.0	70.8	71.8	72.8	73.3	76.1	78.0
	5%	WB	75.9	75.8	74.9	72.1	68.9	64.8	64.4	66.4	68.3	68.7	72.3	74.7
		MCDB	79.4	79.3	78.4	75.8	73.8	70.7	69.8	71.1	72.2	72.2	75.0	76.9
	10%	WB	74.5	74.5	73.7	70.7	67.7	63.3	62.9	65.1	67.0	67.6	70.8	73.4
		MCDB	78.3	78.5	77.5	74.6	72.6	69.6	68.7	70.2	71.3	71.4	73.9	76.2
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.4	8.5	9.8	11.7	13.4	13.1	11.0	9.5	8.4	8.1	7.9
		MCDBR	9.3	9.5	9.8	10.9	13.8	15.8	15.9	12.9	10.8	9.5	9.6	9.3
		MCWBR	6.4	6.3	6.9	7.7	9.6	12.5	12.5	10.4	8.2	7.8	7.3	6.6
	5% WB	MCDBR	9.3	9.0	9.0	10.0	11.4	14.2	14.5	12.0	9.7	9.1	8.7	8.2
		MCWBR	6.7	6.6	6.6	7.4	9.2	12.3	12.1	10.3	8.1	7.4	6.8	6.4
Clear-Sky Solar Irradiance	taub	0.446	0.442	0.430	0.396	0.372	0.354	0.364	0.428	0.470	0.473	0.448	0.447	
	taud	2.327	2.344	2.358	2.417	2.447	2.471	2.430	2.238	2.134	2.171	2.282	2.310	
	Ebn at Noon	285	280	272	264	255	253	255	249	255	267	282	285	
	Edh at Noon	43	41	38	33	28	26	28	38	47	49	45	44	
All-Sky Solar Radiation	RadAvg	1896	1820	1560	1275	1060	948	1014	1199	1406	1568	1754	1840	
	RadStd	161	139	85	85	54	49	66	76	95	146	128	133	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page